Constellation

James A. FitzPatrick NPP
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Alexander Sterio
Site Vice President– JAF

JAFP-24-0061
November 22, 2024

United States Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, D.C. 20555-0001

James A. FitzPatrick Nuclear Power Plant
Renewed Facility Operating License No. DPR-059
NRC Docket No. 50-333

Subject: LER: 2024-003, Turbine Trip and Scram due to Automatic Voltage Regulator Protective Device during a Grid Transient

Dear Sir or Madam:

This report is being submitted pursuant to 10 CFR 50.73(a)(2)(iv)(A).

There are no new regulatory commitments contained in this report.

Questions concerning this report may be addressed to Mr. Mark Hawes, Regulatory Assurance, at (315) 349-6659.

Sincerely,

Alexander Sterio
Site Vice President

ADS/MH

Enclosure: LER: 2024-003, Turbine Trip and Scram due to Automatic Voltage Regulator Protective Device during a Grid Transient

cc: USNRC, Region I Administrator
USNRC, Project Manager
USNRC, Resident Inspector
INPO Records Center (IRIS)

U.S. NUCLEAR REGULATORY COMMISSION

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name		2. Docket Number	3. Page
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4. Title
Turbine Trip and Scram due to Automatic Voltage Regulator Protective Device during a Grid Transient

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name		Docket Number
09	23	2024	2024 -	003 -	00	11	22	2024	N/A	☐ 050	N/A
									N/A	☐ 052	N/A

9. Operating Mode	10. Power Level
1	100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (*Check all that apply*)

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☑ *50.73(a)(2)(iv)(A)*	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify in Abstract below or in NRC Form 366A).

12. Licensee Contact for this LER

Licensee Contact	Telephone Number (Include Area Code)
Mr. Mark Hawes, Regulatory Assurance	315-349-6659

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☑ No	☐ **Yes** (If yes, complete 15. Expected Submission date)				

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

At 0720 on September 23, 2024, James A. FitzPatrick Nuclear Power Plant (JAF) was at 100% power when an automatic scram occurred as a result of a main turbine trip due to an automatic trip of the generator output breakers. All control rods inserted and a subsequent Reactor Pressure Vessel (RPV) low water level resulted in Group 2 isolation and system initiation of High Pressure Coolant Injection (HPCI) and Reactor Core Isolation Cooling (RCIC). RCIC did inject but HPCI did not inject as expected based on RPV water level recovery with the Feedwater System.

The generator output breakers were tripped because of the energization of the digital Automatic Voltage Regulator (AVR) protective device crowbar circuit due to an overvoltage condition during a grid transient. A corrective action is planned to evaluate and model the event to determine if the min limiter function or software changes to the AVR should be implemented.

This event is reportable in accordance with CFR 50.73(a)(2)(iv)(A) as a Reactor Protection System (RPS) actuation, including a general containment Group 2 isolation and HPCI and RCIC system actuation.

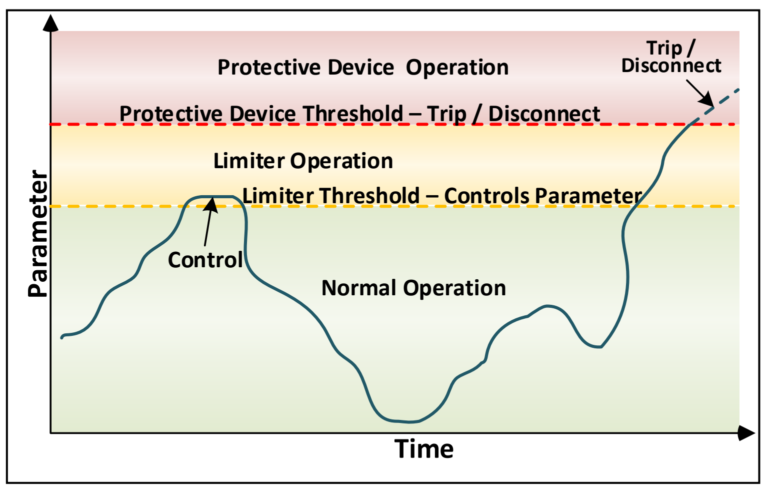
NARRATIVE

Background

The main generator [EIIS identifier:TB] has an Exciter System [TL] to stabilize voltages from the electrical transmission grid. The excitation is controlled by a Digital Automatic Voltage Regulator (AVR). The AVR has two redundant channels.

Reactive Power is measured in units of "mega volt-amperes reactive", or MVAR, and is taken into account when designing and operating power systems. The power associated with reactive power does no net work but current must still be supplied by the conductors, transformers and generators. They must be sized to carry the total current, not just the current that does useful work (active power). Insufficient reactive power can depress voltage levels on an electrical grid and, under certain operating conditions, collapse the network (a blackout).

Therefore, the purpose of the AVR is to control the exciter based on changes on the grid, including reactive power. AVR may enable control parameters such as over excitation limiter (OEL) to prevent overheating from excessive current; and an under excitation limiter (UEL) to prevent de-synchronization from the grid.



Besides control parameters, the AVR is designed with protection devices to trip the system to protect components from being damaged. The protective device associated with this event is the crowbar circuit which is designed to respond to an overvoltage or surge condition.

Event Description

On September 23, 2024, at 0720 with James A. FitzPatrick Nuclear Power Plant (JAF) reactor operating at 100 percent power, an automatic reactor shutdown (scram) occurred because of a main turbine trip due to an automatic trip of the generator output breakers. All control rods inserted and a subsequent Reactor Pressure Vessel (RPV) low water level resulted in Group 2 isolation and system initiation of High Pressure Coolant Injection (HPCI) and Reactor Core Isolation Cooling (RCIC). RCIC did inject but HPCI did not inject as expected based on RPV water level recovery with the Feedwater System.

This event was reported at the time of the event by ENS 573331 and is reportable in accordance with CFR 50.73(a)(2)(iv)(A) as a Reactor Protection System (RPS) actuation, including a general containment Group 2 isolation and HPCI and RCIC system actuation.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
James A. FitzPatrick Nuclear Power Plant	☑ 050 ☐ 052	05000333	2024	– 003	– 00

Event Analysis

The scram event was analyzed based on a transient event originating from the electrical transmission grid. The following describes how electrical systems responded over the course of seconds.

The grid transient started as a large demand for reactive power (MVAR). This caused the AVR to automatically increase reactive power output by increasing exciter field current. When the grid transient ceased, reactive power fell and the AVR automatically decreased exciter field current to limit the reactive power output. A back electromagnetic field (EMF) was produced when field current reached zero amps, and this caused exciter field voltage to spike such that the break over diode (BOD) responded and triggered the protective device crowbar circuit to close on overvoltage. The crowbar circuit overvoltage signal sent a signal to the main generator lockout relays to trip the generator.

The AVR current min limiter used for under excitation (UEL) is not enabled. The limiter would theoretically prevent the AVR protective devices from engaging by not allowing operating parameters to reach protective device limits. It is postulated that this limiter could have aided in preventing this event from occurring (see planned action section).

Cause

The cause of the Main Turbine trip, including RPS scram and system actuations, was the energization of the AVR protective device crowbar circuit due to an overvoltage condition during a grid transient.

Similar Events

None

Corrective Actions

Completed Actions

No equipment deficiencies identified at FitzPatrick, allowing the station to return to full power on 9/25/24.

Planned Actions

To prevent a similar grid transient from causing a protective device from actuating, an evaluation will be conducted to model the event and conclude if the AVR min limiter for under excitation (UEL) or other software enhancements should be implemented.

Safety Significance

There were no actual nuclear consequences. The turbine protection system actuated in response to a grid transient without issue. The RPS responded to the main turbine trip to shutdown the reactor and systems actuated as expected.

References

Issue Report - IR 04803804, AOP-1 Reactor Scram, dated September 23, 2024